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(1) Research Program/Project/Study Title: *Conservation importance of an isolated forest fragment on Panay Island*

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Component Title:	Lowland Key Biodiversity Area
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(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields Natural/Biological Sciences ☐ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
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(7) Sector/Commodity Environmental Science	(8) Discipline/Field of Specialization Biodiversity and Conservation
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(8) Rationale
ABSTRACT

The Critically endangered Rufous-headed hornbill *Rhabdotorrhinus waldeni*, the Endangered Visayan Tarictic Hornbill *Penelopides panini* and nine other threatened birds were observed in the lowland dipterocarp forests of Mt. Agudo Mountain Range in Capiz, Panay Island last 8-12 May 2021. Using point counts, mist netting and interviews, 68 species of birds including a new island record were recorded. We evaluate the importance of the mountain range for bird conservation, in relation to the three other Protected Areas in Panay Island. We propose conservation measures to enhance its importance for bird conservation.

/(9-A) Project Description

Recent taxonomic changes and discoveries of new species have significantly increased the number of birds in the Philippines from 572 to 722 species (Allen 2020; Kennedy et al. 2000). Of the 722 species, 254 (35%) are endemic only to the Philippines and 103 of this are threatened (Allen 2020). A number of the threatened and endemic birds are restricted only in one or a few cluster of islands. Among the group of islands with the highest concentration of threatened endemic birds is the Western Visayas.

The Western Visayas is composed of the islands of Panay, Negros, Guimaras, Cebu, Masbate and Ticao (also called the Negros-Panay faunal region) and is home to 29 species of threatened endemic birds of which, 18 species were known only to this group of islands (Brooks et al. 2000; Paguntalan et al. 2002). Despite the high numbers of threatened endemic wildlife, very few studies had been conducted on Western Visayas particularly the island of Panay (Slade et al. 2005; Klop et al. 2000; Klop et al. 1998).

The destruction of natural habitat and their transformation into agricultural land, plantation and settlement were the two major drivers of the current loss in biodiversity (Jantz et al. 2015; Maxwell et al. 2016). The forests of Panay Island had suffered severe forest loss and what is left of its forests were highly fragmented (Klop et al. 2000; Collar et al. 1999). Three patches of forests had been declared as protected areas: Northwest Panay Peninsula Protected Landscape, Sibalom Natural Park, and the Bulabog-Putian Natural Park. The largest remaining forest of Central Panay Mountain Range currently has no legal protection.

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Other small forests appear to be natural and degraded including the Mt. Agudo Mountain Range (MAMR) in the Province of Capiz.

The MAMR is the highest mountain in Pilar and President Roxas municipalities with elevation reaching 812 m asl. The MAMR serves as a watershed to at least four municipalities surrounding the mountain. The Provincial Environment and Natural Resources Office of the Provincial Government of Capiz is looking into the conservation and protection of the remaining patches of forests in Capiz. We examine the avifaunal community of MAMR and compare the bird diversity with the other protected areas.

(10) Objectives/Statement of the Problem

1. Assess the conservation importance of endemic species and key biodiversity areas.

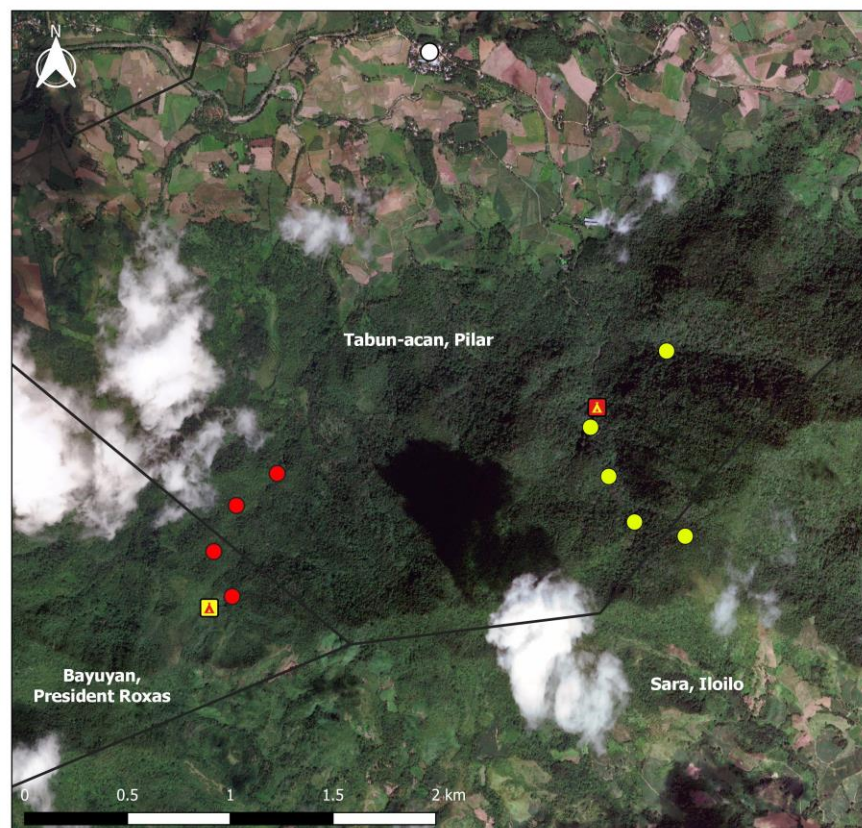
(11) Methodology

Site description

The Mt. Agudo Mountain Range forms an arc from the town of Pilar towards barangay Bayuyan in President Roxas (formerly Lutod-Lutod). Large, tall dipterocarp trees of about 25 meters tall grows along steep gullies and slopes to form patches of tropical lowland rainforest. The rest of the western side of the mountain was covered with advance second growth interspersed with *Cocos nucifera*. We conducted surveys in the secondary forest in Barangays Yating and lowland dipterocarp forest in Tabun-acan (Figure 1).

Figure 1. Location of bird survey areas in Mt. Agudo Mountain Range.

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Bird survey points in Agodo Mt. Range, Pilar, Capiz (May 2021)

WGS84 / Pseudo-Mercator
Scale - 1 : 20,000

Data source: Bing Aerial, CaPENRO & PhilBio

Legend:

- Survey points - Hinulugan
- Survey points - Bayuyan
- ▲ Bayuyan campsite
- ▲ Hinulugan campsite
- Tabun-acan Brgy. hall
- Political boundaries



Mist-netting

We erected 13 monofilament mist-nets (12x2.6 meter with 34 mm mesh size) in the sub-canopy and understory level. Nets were closely monitored at all times with assigned observers moving between nets of at least 30 min to one hour interval depending on the rate of catch. Nets were moved after three days to a new location, preferably close to observed flyways and where there area fruiting and flowering trees.

Bird observations using point transects

We established 12 count stations along trails in advance second growth and mature secondary forests. Each point is set at least 250 meters apart with a maximum radial distance of 200 meters and marked in the field using a Global Positioning System (GPS). A 10x42 and 8x42 roof type binoculars were used in observing birds starting at 0600hrs until 0900hrs. For every positive record, we note sightings or call, number of birds, sex and estimated distance from point of detection. Count duration at each point lasted eight minutes. We use rangefinders in determining point-to-observation perpendicular distance. References used in identifying birds include Kennedy et al. (2000), Jakosalem et al (2019) and Allen (2020). Survey was conducted from 5 to 12 May 2021 in President Roxas and Yating forests in Pilar, Capiz.

Measuring species diversity and similarity

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The Shannon – Weiner index of diversity was used to determining species richness between islands. This index takes both abundance and richness into account using the formula:

$$H = - \sum [p_i \cdot \ln(p_i)] \text{ where}$$

Sum is the summation, p_i is the proportion of each species in a sample. Evenness of the distribution species is calculated by dividing the Shannon's diversity index H by natural logarithm of species richness $\ln(S)$.

(12) Results and Discussion

A total of 68 species of birds were recorded in Mt. Agudo Mountain Range. Of the 68 species, 34 (50%) were endemic to the Philippines, 10 were restricted to the West Visayas region, eight were globally threatened and one was a migratory bird.

The most abundant bird in the lowland dipterocarp forests was the Visayan Bulbul *Ixos guimarasensis* (17%) followed by the Buff-eared Brown Dove *Phapitreron nigrorum* (14%) and the Endangered Visayan Tarictic Hornbill *Penelopides panini* at 7% (Figure 2a). The Elegant Tit *Periparus elegans* shared 5% of the bird proportion with Magnificent Sunbird and Balicassiao. The White-bellied Munia (*Lonchura*), Yellow-vented Bulbul *Pycnonotus goaivier* and the Black-headed Munia (*Lonchura atricapilla*) were the top three most abundant birds in second growth areas (Figure 2b).

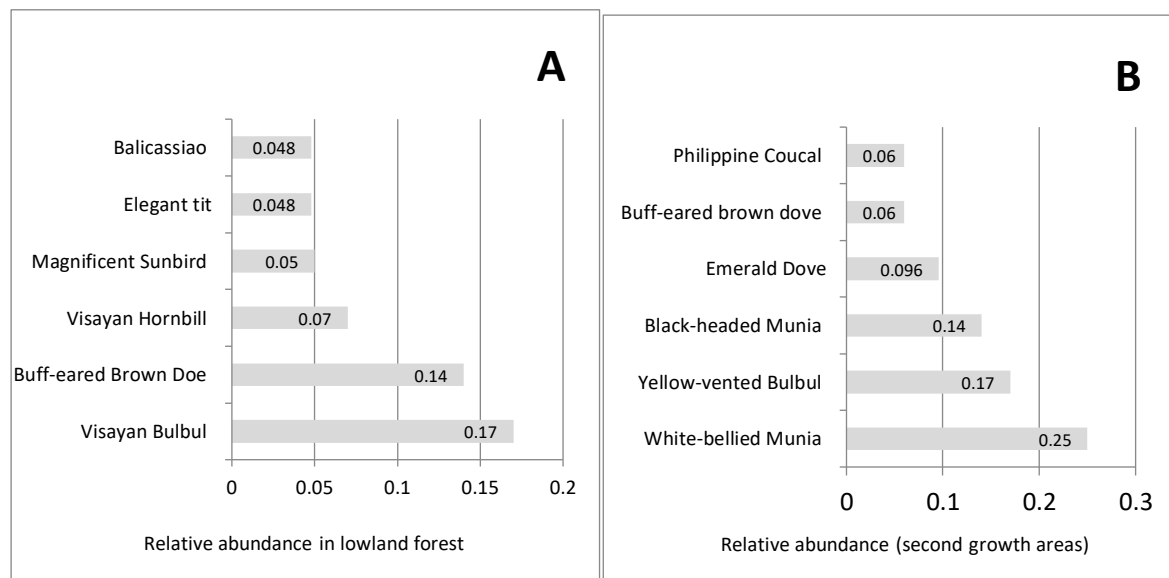


Figure 2. The relative abundance of the top five most abundant birds in second growth and lowland forests of Mt. Agudo Mountain Range.

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In terms of mist-netting, a total of 63 individuals were mist-netted in Mt. Agudo Mountain Range, 63 of which were caught in lowland forest while the rest were in second growth and open areas. The most frequently captured bird was the I. guimarasensis (24), followed by the Philippine Glossy Swiftlet (15), Visayan Drongo (6) and the Buff-eared Brown Dove (4). The rest of the species were represented by less than three individuals (Table 4).

Table 1. Species composition and total number of individuals mist-netted in Mt. Agudo Mountain Range.

Species Name	Forest	Second growth
Buff-eared Brown Dove <i>Phapitreron nigrorum</i>	4	
Grey-capped Emerald Dove <i>Chalcophaps indica</i>	2	
Visayan Bulbul <i>Ixos guimarasensis</i>	24	1
Philippine Glossy Swiftlet <i>Collocalia marginata</i>		15
Collared Kingfisher <i>Todiramphus chloris</i>		2
Balicassiao <i>Dicrurus (balicassius) mirabilis</i>	6	1
Philippine Nightjar <i>Caprimulgus manillensis</i>		1
Philippine Hawk Cuckoo <i>Hierococcyx pectoralis</i>	1	
Chestnut-crowned Tailorbird <i>Orthotomus castaneiceps</i>	2	
Gray's Grasshopper Warbler <i>Locustella fasciolata</i>		1
Philippine Magpie Robin <i>Copsychus mindanensis</i>	1	
Black-naped Monarch <i>Hypothymis azurea</i>		2
Magnificent Sunbird <i>Aethopyga magnifica</i>	2	1
Black-headed Munia <i>Locustella atricapilla</i>		1
Total number of individuals	63	4

Among the interesting find was the capture of the Gray's Grasshopper Warbler *Locustella fasciolata*. This is the first confirmed record of this migratory bird in Panay Island. One individual was caught in a mist-net set along the secondary growth beside a regenerating forest. Gray's Grasshopper Warbler are migratory species passing through Philippines on the months of August to May (Allen 2020; Kennedy et al. 2000).

Threatened and near-threatened species

We recorded nine threatened and near-threatened birds, of which four were classified as Critically Endangered under the list of threatened species of the Department of Environment and Natural Resources (Department Administrative order 2019 – 09), two Endangered birds and two Vulnerable (Table 3). Among the threatened birds, the Endangered Visayan tarictic hornbill has the highest relative abundance while the rest were represented by less than five individuals.

Table 2. Counts and abundance scores (figures inside parentheses) of threatened and near-threatened species in Mt. Agudo Mountain Range.

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Species Name	Relative Abundance	Threatened category	
		DENR DAO	IUCN
Southern Philippine Hawk Eagle <i>Nisaetus pinskeri</i>	R	En	En
Grey-breasted Brown Dove <i>Phapitreron maculipectus</i>	4 (0.015)		Nt
Visayan Tarictic Hornbill <i>Penelopides panini</i>	22 (0.08)	Cr	En
Rufous-headed Hornbill <i>Rhabdotorrhinus waldeni</i>	2 (0.008)	Cr	Cr
Colasisi <i>Loriculus philippensis</i>	R	Cr	
Flame-templed Babbler <i>Dasycrotapha speciosa</i>	1 (0.004)	En	En
Southern Indigo-banded Kingfisher <i>Ceyx cyanopectus</i>	1 (0.004)	Cr	Nt
Visayan Scops owl <i>Otus nigrorum</i>	5	Vu	Vu
Visayan Flowerpecker <i>Dicaeum haematostictum</i>	1 (0.004)	Vu	Vu

The most significant findings was encountering the Critically Endangered Rufous-headed hornbill *Rhabdotorrhinus waldeni*, Endangered Visayan Tarictic Hornbill *Penelopides panini*, Endangered Flame-templed Babbler *Dasycrotapha speciosa* and four near-threatened birds (Grey-breasted Brown Dove *Phapitreron maculipectus*, Visayan Scops Owl *Otus nigrorum*, Visayan Shama *Copsychus superciliaris*, Black-belted Flowerpecker *Dicaeum haematostictum* (Table 2). Other species noteworthy to mention were Southern Indigo-banded Kingfisher *Ceyx nigrirostris* and the Visayan Cuckooshrike *Coracina panayensis*.

SELECTED SPECIES ACCOUNTS

Rufous – headed hornbill *Rhabdotorrhinus waldeni* - Critically Endangered (IUCN and DENR DAO): Two male individuals were seen and heard calling close to a fruiting vine in mature secondary forest in Pilar. One individual was also heard calling from the forested ravines of Mt. Agudo in barangay Yating.

Visayan Tarictic Hornbill *Penelopides panini* - Endangered (IUCN and DENR DAO): At least four separate groups were encountered in Agkawayan and Pilar areas of the Yating – President Roxas Mountain Range. One male Visayan Tarictic Hornbill was observed guarding a nest tree hole on a ridge. A female was observed to enter the nest on 10 May 2021 but left after a minute.

Flame-templed Babbler *Dasycrotapha speciosa* – Endangered (IUCN and DAO): The Flame-templed Babbler was uncommon in both areas with more encounters in Pilar despite the shorter period of sampling. Two individuals were observed searching for insects along herbaceous and woody vines in the understory of a mature secondary forest in Pilar. Individuals encountered in Agkawayan were also found in areas with relatively taller and larger trees.

Southern Indigo-banded Kingfisher *Ceyx nigrirostris* - Near-threatened (IUCN); Critically Endangered (DENR DAO): Two individuals were observed in two separate occasion along the Agkawayan stream. Another individual was observed in one of the tributaries of Hinulogan falls.

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Grey-breasted Brown Dove *Phapitreron maculipectus* - Near-threatened (IUCN); Vulnerable (DENR DAO): The Grey-breasted Brown Dove was observed in both Agkawayan and Hinulugan. The bird was uncommon and often heard calling than seen. One individual was seen briefly feeding on ripe fruits of *Syzigium* in Agkawayan.

Visayan Scops Owl *Otus nigrorum* - Vulnerable (IUCN and DENR DAO): Two individuals were heard calling from opposite direction less than 200 meters from the campsite in Agkawayan. Two other individuals were heard in a dipterocarp forest surrounding Hinulogan Falls. The bird was uncommon in both areas compared to the Luzon hawk owl *Ninox luzonensis*.

(13) Conclusion

Our surveys were conducted within a short period of time and does not include the seasonal changes in distribution and abundance of birds. The limited sample size and coverage only represents the situation in a specific locality within MAMR. The generality of these results should be considered with caution.

The Rufous-headed Hornbill, Visayan Cuckooshrike and Flame-templed Babbler still survives in degraded forest fragments but in lower abundances. This could mean that these species were still able to tolerate to some degree habitat disturbances. The largest and smallest species tend to decrease in relative abundance or become extirpated as forest fragments become reduced in area and with more isolation (Brown and Sullivan 2005). Further fragmentation and degradation of the area would have detrimental impacts on the remaining populations.

There were also species expected to occur in lowland dipterocarp forests that were not encountered during the survey. This include the Visayan Fantail *Rhipidura albiventris*, Sulphur-billed Nuthatch *Sitta oenochlamys*, White-throated Jungle Flycatcher *Rhynomyias albigularis*, White-winged Cuckooshrike *Coracina ostenta*, Philippine Oriole *Oriolus steerii*, and the two woodpeckers (White-bellied Woodpecker *Dryocopus javensis* and Yellow-faced Flameback *Chrysocolaptes xanthocephalus*). It seems likely that reduced forest quality and differences in microclimate (Viana et al. 1997; Saunders et al. 1991) may play a role in declines or losses of the above mentioned birds.

The relative abundance of the forest specialist *P. panini* was one of the highest among the birds observed in MAMR. This species is known to tolerate forest disturbances and survive in badly degraded habitats (del Hoyo et al. 2016; Poonswad et al. 2014; Collar et al. 1999). It is possible that with less than optimum habitat conditions, *P. panini* is forced to adapt to the available environment. The presence of both species of Visayan hornbills may not indicate the preference for this type of habitat by rather lack of alternate choices (Brown and Sullivan 2005).

The Mt. Agudo mountain range was separated from the larger forests of the Central Panay Mountain Range (CPMR). It would also follow that the surviving populations of birds in the area were isolated from the main population in the Central Panay Mountain Range. The

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relative abundance of the species and the absence of expected lowland species already indicates the impact of forest loss on bird abundance and diversity.

Conservation initiatives

The patches of lowland dipterocarp forests lies within the 668 hectares of forestland in Barangay Tabun-acan in Pilar and was awarded by the Department of Environment and Natural Resources (DENR) to the Samahan ng mga Magbubukid ng Tabun-acan Pilar Capiz Association Inc. (SMTPCAI) through a Community-based Forest Management Agreement (CBFMA). The rest of the mountain range was within the forestland.

The SMTPCAI and the Tabun-acan barangay council have significantly played a role in protecting the remaining lowland forests for more than 25 years. The use of native tree species and assisted natural regeneration practices has transformed the Mahogany *Sweetenia macrophylla* and *Gmelina arborea* tree plantations into a secondary lowland forest over time.

(14) Recommendations

MANAGEMENT RECOMMENDATIONS

Tree planting activities using native species of trees should be strategically conducted to facilitate connectivity of forest fragments. The planting of native trees in between forest patches and in clearings would facilitate movement of species or populations between forest patches. The protection and rehabilitation of the remaining forests particularly towards the side bordering Iloilo Province and those areas utilized as pastureland is important in ensuring the long term conservation of the bird diversity in the area.

The conservation and management of the Mt. Agudu Mountain Range must consider the involvement of the municipalities of Pilar and President Roxas and the barangays within the mountain range. The Peoples Organization and the barangay councils have significantly played a role in protecting the remaining forests. The experience of the Peoples Organization in President Roxas demonstrated the important role of the local communities and local government units particularly at the barangay level in protecting the watershed and wildlife in the area. There is a need to build the capacity of the Peoples Organization in managing their Community-based Forest Management Agreement. Mainstreaming biodiversity in the CBFMA Resource Use Plan and the CBFMA Framework Plan as well as updating the forest cover and status using forest ecosystem types were among the most important needs of the area.

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